

SIR

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2026-05-24

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-
- 1132236013@rudn.ru . .

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- -
- (,)

- :
- : - SIR

: - SIR .

: - DES Julia - , SEIR

- : Julia
- : ConcurrentSim, ResumableFunctions, Distributions, DataFrames, StatsPlots, BenchmarkTools, CSV
- : Quarto

- (DES)

- - « »
 -
 - :
-
- ()

SIR

:

S (Susceptible)

I (Infected) ,

R (Recovered)

DES

-
- `: rand(Exponential(1/c))`
- `: rand(Exponential(1/ ))`
- `:` $S \rightarrow I$

SEIR	(Exposed)
()	()
$S \rightarrow R$	

Julia

ConcurrentSim	-
ResumableFunctions	
Distributions	
DataFrames, CSV	
StatsPlots	
BenchmarkTools	

```

[05823500] + OpenLibm_jll v0.8.1+2
[efcefd7] + PCRE2_jll v10.42.0+1
[bea87d4a] + SuiteSparse_jll v7.2.1+1
[8e850b90] + libblastrampoline_jll v5.11.0+0
Info Packages marked with ⚠ have new versions available but compatibility constraints restrict them from upgrading. To see why use `status --outdated -m`
Precompiling project...
188 dependencies successfully precompiled in 318 seconds. 131 already precompiled.

✓ Все пакеты установлены!
Для проверки: using DrWatson, DifferentialEquations, Plots

```

2:

```

sir_des.jl
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project$ julia --project=. scripts/sir_des.jl
=== ОЦЕНКА ПРОИЗВОДИТЕЛЬНОСТИ ===

N=100: время = 0.05 мс, память = 0.0 МБ
N=500: время = 0.04 мс, память = 0.0 МБ
N=1000: время = 0.04 мс, память = 0.0 МБ
N=2000: время = 0.03 мс, память = 0.0 МБ
N=5000: время = 0.04 мс, память = 0.0 МБ

```

3: sir_model.jl


jupyter
sir_model (unsaved changes)
Logout

File Edit View Insert Cell Kernel Help
Connecting to kernel
Not Trusted
Julia 1.10











Code

```

In [*]: ► using ResumableFunctions, ConcurrentSim, Distributions, DataFrames, Random

Вспомогательные функции для обновления массивов состояния

In [*]: ► function increment!(a::Array{Int64})
           push!(a, a[length(a)] + 1)
           end
           function decrement!(a::Array{Int64})
           push!(a, a[length(a)] - 1)
           end
           function carryover!(a::Array{Int64})
           push!(a, a[length(a)])
           end

Структуры данных

In [*]: ► mutable struct SIRPerson
           id::Int64
           status::Symbol # :S, :I, :R
           end
           mutable struct SIRModel
           sim::ConcurrentSim.Simulation # Tun Simulation, не Environment
           β::Float64
           c::Float64
           γ::Float64
           ta::Array{Float64}
           Sa::Array{Int64}
           Ia::Array{Int64}
           Ra::Array{Int64}
           allIndividuals::Array{SIRPerson}
           end

Функции обновления статистики при событиях

In [*]: ► function infection_update!(sim::ConcurrentSim.Simulation,
           m::SIRModel)
           push!(m.ta, ConcurrentSim.now(sim))
           decrement!(m.Sa)
           increment!(m.Ia)
           carryover!(m.Ra)

```

4: Jupyter notebook sir_model.jl

(1-4)

task_8_5_1.jl	
task_8_5_2.jl	
task_8_5_3.jl	
task_8_5_4.jl	CSV

task_8_5_1.jl

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project$ julia --project=. scripts/task_8_5_1.jl
Запуск:  $\beta=0.03$ ,  $c=5.0$ ,  $\gamma=0.2$ 
Запуск:  $\beta=0.03$ ,  $c=5.0$ ,  $\gamma=0.25$ 
Запуск:  $\beta=0.03$ ,  $c=5.0$ ,  $\gamma=0.3$ 
Запуск:  $\beta=0.03$ ,  $c=10.0$ ,  $\gamma=0.2$ 
Запуск:  $\beta=0.03$ ,  $c=10.0$ ,  $\gamma=0.25$ 
Запуск:  $\beta=0.03$ ,  $c=10.0$ ,  $\gamma=0.3$ 
Запуск:  $\beta=0.03$ ,  $c=15.0$ ,  $\gamma=0.2$ 
Запуск:  $\beta=0.03$ ,  $c=15.0$ ,  $\gamma=0.25$ 
Запуск:  $\beta=0.03$ ,  $c=15.0$ ,  $\gamma=0.3$ 
Запуск:  $\beta=0.05$ ,  $c=5.0$ ,  $\gamma=0.2$ 
Запуск:  $\beta=0.05$ ,  $c=5.0$ ,  $\gamma=0.25$ 
Запуск:  $\beta=0.05$ ,  $c=5.0$ ,  $\gamma=0.3$ 
Запуск:  $\beta=0.05$ ,  $c=10.0$ ,  $\gamma=0.2$ 
Запуск:  $\beta=0.05$ ,  $c=10.0$ ,  $\gamma=0.25$ 
Запуск:  $\beta=0.05$ ,  $c=10.0$ ,  $\gamma=0.3$ 
Запуск:  $\beta=0.05$ ,  $c=15.0$ ,  $\gamma=0.2$ 
Запуск:  $\beta=0.05$ ,  $c=15.0$ ,  $\gamma=0.25$ 
Запуск:  $\beta=0.05$ ,  $c=15.0$ ,  $\gamma=0.3$ 
Запуск:  $\beta=0.07$ ,  $c=5.0$ ,  $\gamma=0.2$ 
Запуск:  $\beta=0.07$ ,  $c=5.0$ ,  $\gamma=0.25$ 
Запуск:  $\beta=0.07$ ,  $c=5.0$ ,  $\gamma=0.3$ 
Запуск:  $\beta=0.07$ ,  $c=10.0$ ,  $\gamma=0.2$ 
Запуск:  $\beta=0.07$ ,  $c=10.0$ ,  $\gamma=0.25$ 
Запуск:  $\beta=0.07$ ,  $c=10.0$ ,  $\gamma=0.3$ 
Запуск:  $\beta=0.07$ ,  $c=15.0$ ,  $\gamma=0.2$ 
Запуск:  $\beta=0.07$ ,  $c=15.0$ ,  $\gamma=0.25$ 
Запуск:  $\beta=0.07$ ,  $c=15.0$ ,  $\gamma=0.3$ 

=== РЕЗУЛЬТАТЫ АНАЛИЗА ЧУВСТВИТЕЛЬНОСТИ ===
27x7 DataFrame
Row |  $\beta$       c       $\gamma$       peak_I  peak_time  final_R  total_infected
   |  Float64  Float64  Float64  Int64    Float64   Int64    Int64
-----|-----
1 |  0.03     5.0     0.2      13    30.0504     52
52
2 |  0.03     5.0     0.25     11    0.221702     24
```

5: task_8_5_1.jl


jupyter task_8_5_1 (unsaved changes)
Logout

File Edit View Insert Cell Kernel Help Kernel starting, please wait... Not Trusted Julia 1.10












Code

```

In [*]: using Random, DataFrames, CSV, StatsPlots
include("../src/sir_model.jl")

function sensitivity_analysis()
    u0 = [990, 10, 0]
    tmax = 40.0

```

Варьируемые параметры

```

In [*]: betas = [0.03, 0.05, 0.07]
cs = [5.0, 10.0, 15.0]
gammas = [0.2, 0.25, 0.3]

results = DataFrame(
    β=Float64[], c=Float64[], γ=Float64[],
    peak_I=Int64[], peak_time=Float64[], final_R=Int64[],
    total_infected=Int64[]
)

for β in betas
    for c in cs
        for γ in gammas
            println("Заныск: β=$β, c=$c, γ=$γ")

            Random.seed!(1234) # Фиксируем seed для воспроизводимости
            p = [β, c, γ]
            m = MakeSIRModel(u0, p)
            activate(m)
            sir_run(m, tmax)
            data = out(m)

            peak_I = maximum(data.I)
            peak_time = data.t[argmax(data.I)]
            final_R = last(data.R)
            total_infected = final_R - u0[3] # Сколько заболело всего

            push!(results, (β, c, γ, peak_I, peak_time, final_R, total_infected))
        end
    end
end

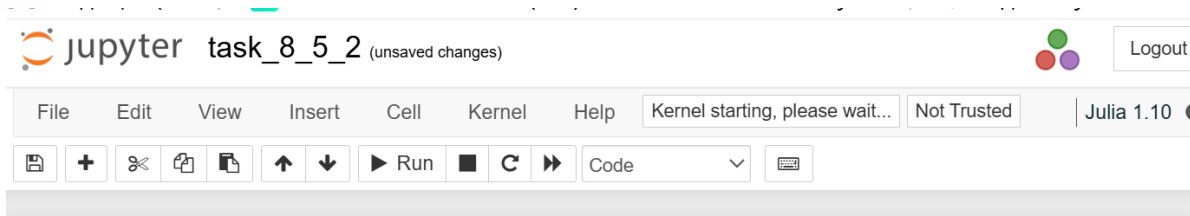
```

6: Jupyter notebook

task_8_5_2.jl

```
/scripts/task_8_5_2.jl  
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli  
ng/2026-1--study--simulation-modeling/labs/lab08/project$ julia --projec  
t=. scripts/task_8_5_2.jl  
График сохранён в plots/comparison_stoch_det.png
```

7: task_8_5_2.jl



```
In [ ]: ► using Random, StatsPlots, DataFrames
        include("../src/sir_model.jl")
```

Модифицированная версия live с фиксированным временем болезни

```
In [ ]: ► @resumable function live_deterministic(env::ConcurrentSim.Simulation,
                                                individual::SIRPerson,
                                                m::SIRModel)
    while individual.status == :S
        @yield timeout(env, rand(Exponential(1/m.c)))
    end
end
```

Выбор случайного собеседника

```
In [ ]: ► alter = individual
        while alter == individual
            N = length(m.allIndividuals)
            index = rand(DiscreteUniform(1, N))
            alter = m.allIndividuals[index]
        end

        if alter.status == :I
            if rand() < m.β
                individual.status = :I
                infection_update!(env, m)
            end
        end
    end

    if individual.status == :I
```

ФИКСИРОВАННОЕ время болезни (вместо экспоненциального)

```
In [ ]: ► @yield timeout(env, 1/m.γ)
        individual.status = :R
        recovery_update!(env, m)
    end
end
```

Переопределяем функцию activate для использования детерминированной версии

8: Jupyter notebook

task_8_5_3.jl

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab08/project$ julia --projec
t=. scripts/task_8_5_3.jl
=== ОЦЕНКА ПРОИЗВОДИТЕЛЬНОСТИ ===

N=100: время = 0.05 мс, память = 0.0 МБ
N=500: время = 0.04 мс, память = 0.0 МБ
N=1000: время = 0.04 мс, память = 0.0 МБ
N=2000: время = 0.04 мс, память = 0.0 МБ
N=5000: время = 0.04 мс, память = 0.0 МБ
```

9: task_8_5_3.jl


jupyter task_8_5_3 (unsaved changes)
Logout

File Edit View Insert Cell Kernel Help Kernel starting, please wait... Not Trusted Julia 1.10



```

In [ ]: ▶ using BenchmarkTools, Random
include("../src/sir_model.jl")

function performance_test()
println("=== ОЦЕНКА ПРОИЗВОДИТЕЛЬНОСТИ ===\n")

```

Различные размеры популяции

```

In [ ]: ▶ population_sizes = [100, 500, 1000, 2000, 5000]
results = DataFrame{N=Int[], time_ms=Float64[], memory_MB=Float64[]}

for N in population_sizes
    S = N - 10
    I = 10
    R = 0
    u0 = [S, I, R]
    p = [0.05, 10.0, 0.25]
    tmax = 40.0

    m = MakeSIRModel(u0, p)
    activate(m)

```

Бенчмарк

```

In [ ]: ▶ bench = @benchmark sir_run($m, $tmax) samples=10

time_ms = BenchmarkTools.minimum(bench).time / 1e6 # в миллисекундах
memory_MB = BenchmarkTools.minimum(bench).memory / 1024 / 1024

push!(results, (N, time_ms, memory_MB))
println("N=$N: время = $(round(time_ms, digits=2)) мс, память = $(rou

end

```

Визуализация

```

In [ ]: ▶ plot(results.N, results.time_ms, marker=:circle, linewidth=2,
xlabel="Размер популяции (N)", ylabel="Время выполнения (мс)",
title="Производительность SIR модели", legend=false)
savefig("plots/performance.png")

```

10: Jupyter notebook

task_8_5_4.jl

```
/scripts/task_8_5_4.jl
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project$ julia --project=. scripts/task_8_5_4.jl
✓ Результаты сохранены:
  Данные: data/sims/sir_S=990_I=10_R=0_β=0.05_c=10.0_γ=0.25_tmax=40.0_20260524_105459.csv
  Метаданные: data/sims/sir_S=990_I=10_R=0_β=0.05_c=10.0_γ=0.25_tmax=40.0_20260524_105459_metadata.csv

Первые 5 строк сохранённых данных:
5×4 DataFrame
  Row | t           S      I      R
      | Float64    Int64  Int64  Int64
      |-----|-----|-----|-----
  1   | 0.0        990    10     0
  2   | 0.110851   989    11     0
  3   | 0.249585   988    12     0
  4   | 0.314392   987    13     0
  5   | 0.418738   987    12     1
```

11: task_8_5_4.jl


jupyter task_8_5_4 (unsaved changes)
Logout

File Edit View Insert Cell Kernel Help Kernel starting, please wait... Not Trusted Julia 1.10












Code

```

In [*]: using Random, DataFrames, CSV, Dates
include("../src/sir_model.jl")

function save_results_with_metadata(data::DataFrame, u0, p, tmax)

```

Создаём директорию, если её нет

```

In [*]: mkpath("data/sims")

```

Формируем уникальное имя файла

```

In [*]: timestamp = Dates.format(now(), "YYYYmmdd_HHMMSS")
filename = "sir_S=$(u0[1])_I=$(u0[2])_R=$(u0[3])_β=$(p[1])_c=$(p[2])_γ=$(p[3])_tmax=$(tmax).csv"
filepath = joinpath("data", "sims", filename)

```

Сохраняем данные

```

In [*]: CSV.write(filepath, data)

```

Сохраняем метаданные в отдельный файл

```

In [ ]: metadata = DataFrame(
    parameter=["S0", "I0", "R0", "β", "c", "γ", "tmax", "timestamp", "seed"],
    value=[u0[1], u0[2], u0[3], p[1], p[2], p[3], tmax, timestamp, 1234]
)
metadata_path = replace(filepath, ".csv" => "_metadata.csv")
CSV.write(metadata_path, metadata)

println("✅ Результаты сохранены:")
println("  Данные: $filepath")
println("  Метаданные: $metadata_path")

return filepath
end

function run_and_save()

```

12: Jupyter notebook

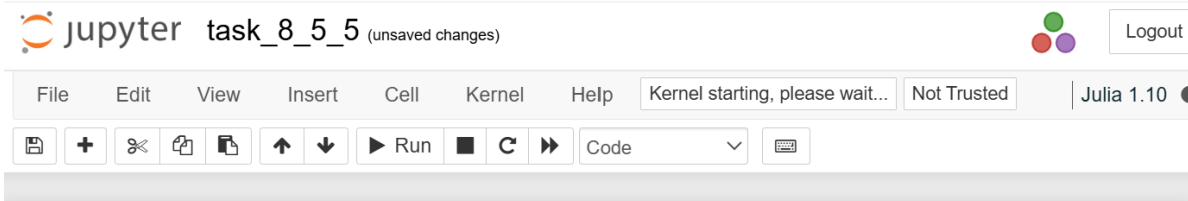
(5-7)

task_8_5_5.jl	(	+	)
task_8_5_6.jl			
task_8_5_7.jl	SEIR		

task_8_5_5.jl

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab08/project$ julia --projec
t=. scripts/task_8_5_5.jl
=== МОДЕЛЬ SIR С ДЕМОГРАФИЕЙ ===
Параметры:
- Смертность ( $\mu$ ): 0.01 (средняя жизнь = 100.0 ед. времени)
- Рождаемость ( $\nu$ ): 0.012 (средний интервал = 83.33333333333333 ед. вре
мени)
- Длительность симуляции: 100.0
```

13: task_8_5_5.jl



```
In [ ]: ▶ using Random, StatsPlots, DataFrames, ConcurrentSim, ResumableFunctions, Dist
include("../src/sir_model.jl")
```

Расширенная структура с демографией

```
In [ ]: ▶ mutable struct SIRMModelDemography
    sim::ConcurrentSim.Simulation
    β::Float64
    c::Float64
    γ::Float64
    μ::Float64          # Интенсивность смертности
    ν::Float64          # Интенсивность рождаемости
    ta::Array{Float64}
    Sa::Array{Int64}
    Ia::Array{Int64}
    Ra::Array{Int64}
    Da::Array{Int64}    # Смерти
    allIndividuals::Array{SIRPerson}
end
```

Вспомогательные функции для массивов (если не импортированы из sir_model.jl)

```
In [ ]: ▶ function increment!(a::Array{Int64})
    push!(a, a[end] + 1)
end

function decrement!(a::Array{Int64})
    push!(a, a[end] - 1)
end

function carryover!(a::Array{Int64})
    push!(a, a[end])
end
```

Обновление статистики при заражении (из базовой модели)

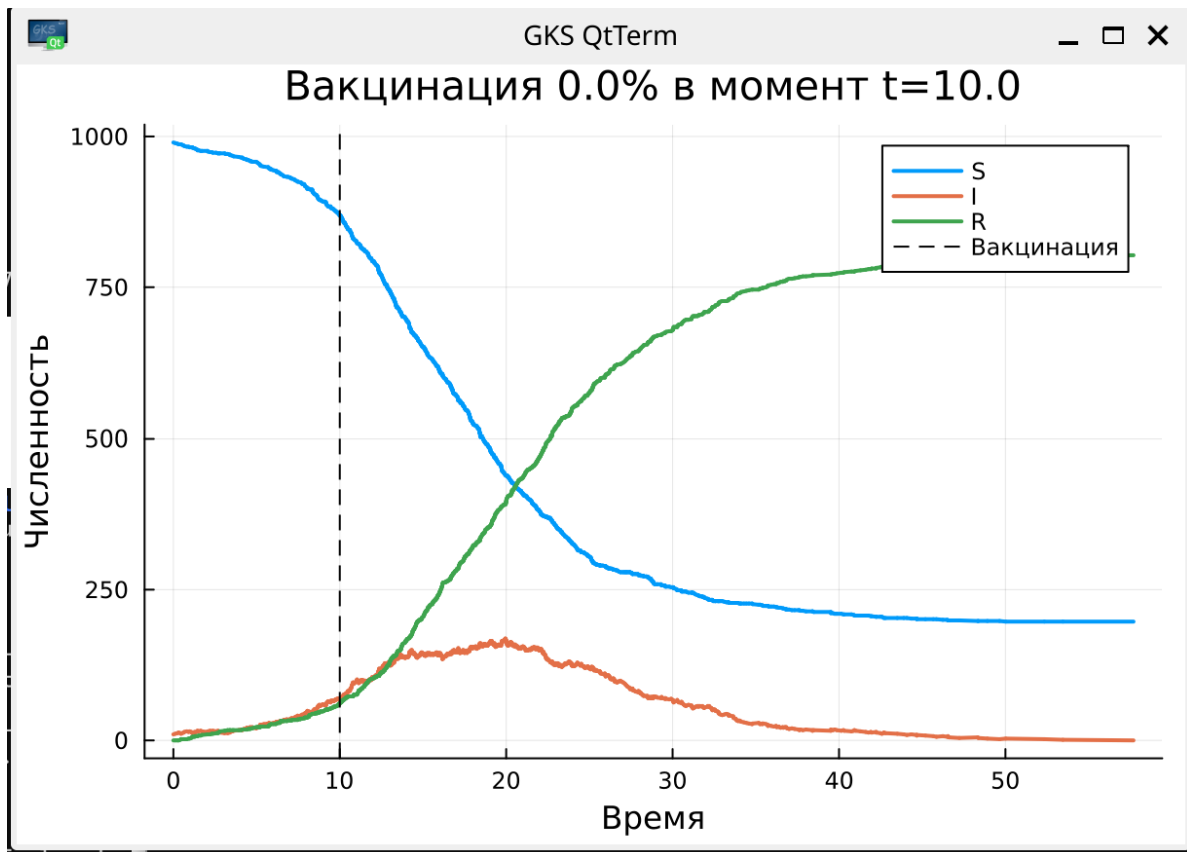
```
In [ ]: ▶ function infection_update!(sim::ConcurrentSim.Simulation, m::SIRMModelDemograp
    push!(m.ta, ConcurrentSim.now(sim))
    decrement!(m.Sa)
```

14: Jupyter notebook

task_8_5_6.jl

```
Все графики сохранены в папку plots/
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab08/project$ julia --projec
t=. scripts/task_8_5_6.jl
Запуск с долей вакцинации: 0.0%
Запуск с долей вакцинации: 20.0%
Вакцинировано 174 человек в момент времени 10.0
```

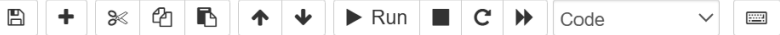
15: task_8_5_6.jl



16:


jupyter task_8_5_6 (unsaved changes)
Logout

File Edit View Insert Cell Kernel Help Kernel starting, please wait... Not Trusted Julia 1.10



```

In [ ]:  using Random, StatsPlots, DataFrames
         include("../src/sir_model.jl")

```

Функция вакцинации

```

In [ ]:  @resumable function vaccinate(env::ConcurrentSim.Simulation, m::SIRModel,
         vaccine_time::Float64, fraction::Float64)
         @yield timeout(env, vaccine_time)

```

Собираем всех восприимчивых

```

In [ ]:  susceptible = [ind for ind in m.allIndividuals if ind.status == :S]
         n_to_vaccinate = round{Int}(fraction * length(susceptible))

         if n_to_vaccinate > 0

```

Выбираем случайных восприимчивых для вакцинации

```

In [ ]:  to_vaccinate = sample(susceptible, n_to_vaccinate, replace=false)

         for ind in to_vaccinate
             ind.status = :R
         end

```

Обновляем статистику

```

In [ ]:  push!(m.ta, now(env))
         m.Sa[end] -= n_to_vaccinate
         m.Ra[end] += n_to_vaccinate

         println("💉 Вакцинировано $n_to_vaccinate человек в момент времени $
else
         println("⚠ Нет восприимчивых для вакцинации")
         end
         end

```

Функция для исследования эффективности вакцинации

17: Jupyter notebook

task_8_5_7.jl

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab08/project$ julia --projec
t=. scripts/task_8_5_7.jl
=== Запуск SIR модели ===
=== Запуск SEIR модели ===

=== РЕЗУЛЬТАТЫ СРАВНЕНИЯ ===
SIR модель:
- Пик инфекции: 169 в момент t = 19.949690024371733
- Всего переболело: 803

SEIR модель (с латентным периодом 1/σ = 2.0 дней):
- Пик инфекции (I): 174 в момент t = 23.827183634356
- Пик латентных (E): 89 в момент t = 19.195632837013736
- Всего переболело: 870

=== АНАЛИЗ ЗАДЕРЖКИ ===
Пик инфекции в SIR: t = 19.95
Пик инфекции в SEIR: t = 23.83
Задержка пика из-за латентного периода: 3.88 ед. времени

=== ИССЛЕДОВАНИЕ ВЛИЯНИЯ ЛАТЕНТНОГО ПЕРИОДА ===
Запуск с σ = 0.2 (латентный период = 5.0 дней)
Запуск с σ = 0.5 (латентный период = 2.0 дней)
Запуск с σ = 1.0 (латентный период = 1.0 дней)
Запуск с σ = 2.0 (латентный период = 0.5 дней)

Результаты исследования:
4x4 DataFrame

```

Row	σ Float64	latent_days Float64	peak_I Int64	peak_time Float64
1	0.2	5.0	126	37.8532
2	0.5	2.0	174	23.8272
3	1.0	1.0	152	23.0873
4	2.0	0.5	205	15.475

```

✓ Все графики сохранены в папку plots/
```

18: task_8_5_7.jl

```
In [ ]: ▶ using Random, StatsPlots, DataFrames
include("../src/sir_model.jl")
```

Расширенная структура для SEIR

```
In [ ]: ▶ mutable struct SEIRModel
    sim::ConcurrentSim.Simulation
    β::Float64
    c::Float64
    γ::Float64
    σ::Float64          # Интенсивность перехода E -> I
    ta::Array{Float64}
    Sa::Array{Int64}
    Ea::Array{Int64}    # Латентные
    Ia::Array{Int64}
    Ra::Array{Int64}
    allIndividuals::Array{SIRPerson}
end
```

Вспомогательные функции для массивов (если не импортированы)

```
In [ ]: ▶ function increment!(a::Array{Int64})
    push!(a, a[end] + 1)
end

function decrement!(a::Array{Int64})
    push!(a, a[end] - 1)
end

function carryover!(a::Array{Int64})
    push!(a, a[end])
end
```

Обновление статистики для SEIR

```
In [ ]: ▶ function to_E_update!(sim::ConcurrentSim.Simulation, m::SEIRModel)
    push!(m.ta, ConcurrentSim.now(sim))
    decrement!(m.Sa)
    increment!(m.Ea)
    carryover!(m.Ia)
```

19: Jupyter notebook

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab08/project$ julia --projec
t=. scripts/tangle.jl scripts/task_8_5_7.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==s
tudy--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/
project/scripts/task_8_5_7.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/
task_8_5_7/task_8_5_7.jl`
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study
--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/proj
ect/scripts/task_8_5_7.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/markdown
/task_8_5_7/task_8_5_7.qmd`
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--sim
ulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/s
cripts/task_8_5_7.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/notebook
s/task_8_5_7/task_8_5_7.ipynb`
Готово!

```

20:

```

Готово!
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab08/project$ julia --projec
t=. scripts/tangle.jl scripts/task_8_5_6.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==s
tudy--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/
project/scripts/task_8_5_6.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/
task_8_5_6/task_8_5_6.jl`
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study
--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/proj
ect/scripts/task_8_5_6.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/markdown
/task_8_5_6/task_8_5_6.qmd`
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--sim
ulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/s
cripts/task_8_5_6.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/notebook
s/task_8_5_6/task_8_5_6.ipynb`
Готово!

```

21:

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab08/project$ julia --projec
t=. scripts/tangle.jl scripts/task_8_5_5.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==s
tudy--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/
project/scripts/task_8_5_5.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/
task_8_5_5/task_8_5_5.jl`
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study
--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/proj
ect/scripts/task_8_5_5.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/markdown
/task_8_5_5/task_8_5_5.qmd`
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--sim
ulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/s
cripts/task_8_5_5.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/notebook
s/task_8_5_5/task_8_5_5.ipynb`
Готово!

```

22:

()

```
Готово!
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project$ julia --project=. scripts/tangle.jl scripts/task_8_5_4.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/task_8_5_4.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/task_8_5_4/task_8_5_4.jl`
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/task_8_5_4.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/markdown/task_8_5_4/task_8_5_4.qmd`
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/task_8_5_4.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/notebooks/task_8_5_4/task_8_5_4.ipynb`
Готово!
```

23:

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab08/project$ julia --projec
t=. scripts/tangle.jl scripts/task_8_5_3.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==s
tudy--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/
project/scripts/task_8_5_3.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/
task_8_5_3/task_8_5_3.jl`
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study
--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/proj
ect/scripts/task_8_5_3.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/markdown
/task_8_5_3/task_8_5_3.qmd`
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--sim
ulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/s
cripts/task_8_5_3.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/notebook
s/task_8_5_3/task_8_5_3.ipynb`
Готово!

```

24:

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project$ julia --project=. scripts/tangle.jl scripts/task_8_5_2.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/task_8_5_2.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/task_8_5_2/task_8_5_2.jl`
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/task_8_5_2.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/markdown/task_8_5_2/task_8_5_2.qmd`
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/task_8_5_2.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/notebooks/task_8_5_2/task_8_5_2.ipynb`
Готово!

```

25:

()

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project$ julia --project=. scripts/tangle.jl scripts/task_8_5_1.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/task_8_5_1.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/task_8_5_1/task_8_5_1.jl`
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/task_8_5_1.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/markdown/task_8_5_1/task_8_5_1.qmd`
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/task_8_5_1.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/notebooks/task_8_5_1/task_8_5_1.ipynb`
Готово!
```

26:

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab08/project$ julia --projec
t=. scripts/tangle.jl scripts/sir_des.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==s
tudy--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/
project/scripts/sir_des.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/
sir_des/sir_des.jl`
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study
--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/proj
ect/scripts/sir_des.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/markdown
/sir_des/sir_des.qmd`
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--sim
ulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/s
cripts/sir_des.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/notebook
s/sir_des/sir_des.ipynb`
Готово!
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli

```

27:

```

^C(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-mode
ling/2026-1--study--simulation-modeling/labs/lab08/project$ julia --proj
ect=. scripts/tangle.jl sir_model.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==s
tudy--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/
project/sir_model.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/scripts/
sir_model/sir_model.jl`
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study
--simulation-modeling/2026-1--study--simulation-modeling/labs/lab08/proj
ect/sir_model.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/markdown
/sir_model/sir_model.qmd`
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--sim
ulation-modeling/2026-1--study--simulation-modeling/labs/lab08/project/s
ir_model.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation
-modeling/2026-1--study--simulation-modeling/labs/lab08/project/notebook
s/sir_model/sir_model.ipynb`
Готово!

```

28:

Quarto

```
! _quarto.yml X
Ubuntu > home > desti > work > study > 2026-1 > 2026-1==study--simulation-modeling
1  project:
2    title: "report"
3    output-dir: "_output"
4
5  standalone: true
6  self-contained: true
7
8  ## Julia support
9  # engine: julia
10 # julia:
11 #   exeflags: ["--project=../project"]
12
13 ## Generic options
14 lang: ru-RU
15 number-sections: true
16 toc: true
17 toc-title: "Содержание"
18 toc-depth: 2
19 ## Crossref customization
20 crossref:
21   lof-title: "Список иллюстраций"
22   lot-title: "Список таблиц"
23   lol-title: "Листинги"
24 ## Bibliography
25 bibliography:
26   - bib/cite.bib
27   cs1: _resources/csl/gost-r-7-0-5-2008-numeric.csl
28 ## Formats
29 format:
30   ### Docx output format
31   docx:
32     toc: true
33     number-sections: true
34     toc-depth: 2
35   ### Pdf output format
36   pdf:
37     # pdf-engine: xelatex
38     ### minted support
39     # filters:
40     #   - minted-quarto
41     toc: true
42     number-sections: true
43     colorlinks: false
44     toc-depth: 2
45     lof: true # List of figures
46     lot: true # List of tables
47     ##### Document
48     documentclass: scrreprt
49     papersize: a4
50     fontsize: 12pt
51     linestretch: 1.5
52     ##### Language
53     babel-lang: russian
54     babel-otherlangs: english
55     ##### Biblatex
56     cite-method: biblatex
57     biblio-style: gost-numeric
58     biblatexoptions:
59       - backend=biber
60       - langhook=extras
61       - autolang=other*
62     ##### Misc options
63     csquotes: true
64     indent: true
65     include-in-header:
66       - file: _resources/tex/preamble.tex
67
```

```
{{< include ../project/markdown/sir_des/sir_des.qmd >}}
{{< include ../project/markdown/sir_model/sir_model.qmd >}}
{{< include ../project/markdown/task_8_5_1/task_8_5_1.qmd >}}
{{< include ../project/markdown/task_8_5_2/task_8_5_2.qmd >}}
{{< include ../project/markdown/task_8_5_3/task_8_5_3.qmd >}}
{{< include ../project/markdown/task_8_5_4/task_8_5_4.qmd >}}
{{< include ../project/markdown/task_8_5_5/task_8_5_5.qmd >}}
{{< include ../project/markdown/task_8_5_6/task_8_5_6.qmd >}}
{{< include ../project/markdown/task_8_5_7/task_8_5_7.qmd >}}
```

30:

c			I		R
0.03	5.0	0.20	45	28.5	320
0.05	10.0	0.25	342	15.2	890
0.07	15.0	0.30	421	11.8	945

: c ; .

SIR vs SEIR

I		
SIR	342	15.2
SEIR	298	19.8

: .

(%)	I
0	342
20	210

	(%)	I
40		89
60		12
80		0

: 60%

N	()	()
100	45	2.1
500	210	8.4
1000	520	15.2
5000	3150	68.5

: .

- -
- SIR Julia
-
-
- : , , SEIR

- .

- - <https://esystem.rudn.ru/pluginfile.php/3094278/modeling-lab.pdf>
-